I Orbit Velocity Vector

Start with $V = \omega r$

If we are on orbit, then just use time derivative of rotatation

$$R(\theta) = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix} \qquad \hat{q}(t) = R(\theta(t))\,\hat{q}(t_0)$$

$$\dot{\hat{q}} = \underbrace{\hat{\omega}\hat{q} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}\hat{q}}_{\text{To stay on a circle}}$$

Use diagonals to converge

$$Y = k(r^2 - \hat{q}^T\hat{q}) = k(r^2 - \|q - q_c\|^2)$$

$$\Rightarrow g = \begin{bmatrix} \gamma & -\omega \\ \omega & \gamma \end{bmatrix}(q - q_c)$$ To converge & stay on an orbit.

II X-axis Following Vector Field

Distance to line: $d = [0 \;\; 1]\,q$ → Extract y

Have desired line angle go transition from heading straight towards line to moving along line

$$\psi_d = \frac{\pi}{2}\left(1 - \frac{2}{1 + e^{-d/\sigma^2}}\right)$$

σ: slope of sigmoid

$$\Rightarrow g = \begin{bmatrix} \cos(\psi_d) \\ \sin(\psi_d) \end{bmatrix} V_d$$

III Arbitrary Line Following Vector Field

Define a line

q_l: Point on the line in original frame

ψ_l: Orientation of the line

l: line frame c: inertial (robot) frame

$$ {}^{c}R = \begin{bmatrix} \cos(\psi_l) & -\sin(\psi_l) \\ \sin(\psi_l) & \cos(\psi_l) \end{bmatrix} $$

Rotate from line frame to inertial

$$ {}^{l}c R = {}^{c}R^{T} $$

Point on line in c-frame

$$ \Rightarrow {}^{l}q = {}^{l}cR\,({}^{c}q - {}^{c}q_l) $$

$$ \Rightarrow {}^{l}g = V_d \begin{bmatrix} c\psi_d \\ s\psi_d \end{bmatrix}, \quad {}^{c}g = {}^{c}R \cdot g_l = V_d\, {}^{c}R \begin{bmatrix} c\psi_d \\ s\psi_d \end{bmatrix} $$

$$ g = V_d \cdot {}^{c}R \begin{bmatrix} \cos\psi_d \\ \sin\psi_d \end{bmatrix} $$

$$ q = {}^{l}c R\,({}^{c}q - q_l) $$

$$ d = [0 \quad 1]\, q $$

q_l: point on line in original frame

ψ_l: orientation of the line